

Tan

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EDUCATION

University of California, San Diego

Expected: 12/2025

Master of Science in Data Science

GPA: 3.7/4.0

Courses: Machine Learning (Linear Regression, Logistic Regression, Decision Trees, Random Forest, K-means, SVM, KNN, etc.), Deep Learning (CNNs, RNNs, Transformers, etc.), NLP (Text Preprocessing, BoW, TF-IDF, Word2Vec), Recommender System

University of California, Santa Cruz

09/2019 - 03/2023

Bachelor of Science in Mathematics, Minor in Computer Science

GPA: 3.7/4.0

SKILLS

Programming: Python (TensorFlow, PyTorch, Scikit-learn, Keras, NLTK, SpaCy), SQL, R, HTML, MATLAB, C/C++

Technical Tools: AWS (S3), Git, IntelliJ, Jupyter Notebook, Power BI

INTERNSHIPS

Docta.ai

San Jose, California

Data Scientist Intern

06/2024 - 09/2024

- Developed a synthetic data generator in **Python** to create low-quality text samples (e.g., footers, repetitions, incomplete sentences) and fine-tuned a **BERT** model, achieving an F-1 score of ~90% while enhancing data preprocessing workflows
- Engineered and deployed web crawlers for data collection, cleaning, and normalization of 100+ million records, using **AWS S3** for dataset storage and improving data quality assurance for dialogue systems and LLM training
- Leveraged advanced data mining techniques to optimize **Datatrove**, a low-quality text filter for processing foreign text, improving its filtering capabilities by 15% and enhancing the preparation of high-quality datasets for downstream tasks

Beijing 01.AI Science and Technology Ltd.

Beijing, China

Model Risk Management Intern

07/2023 - 09/2023

- Built safety-related benchmarks and designed tagging systems for training data used in **LLMs**, ensuring high-quality datasets and reliable model outputs, successfully launching a chatbot platform (**PopAI**) to demonstrate practical applications of these efforts
- Cleaned and processed pre-training data with **Python** from open-source repositories, leveraging **PowerBI** for data visualization to identify inconsistencies and optimize dataset quality, structured the cleaned datasets into a scalable database to enhance feature engineering for downstream tasks, supporting efficient fine-tuning (**SFT**) processes
- Collaborated with stakeholders to present insights on data preparation workflows and visual findings, showcasing the impact on model performance and driving improvements in model safety and reliability

Beijing Chuangcache Technology Company

Beijing, China

Research Intern

06/2021 - 08/2021

- Designed and implemented an anomaly detection system to monitor bandwidth speed in Content Delivery Network servers, utilizing data collection and mining techniques to enhance network performance and identify potential service disruptions
- Processed and cleaned large datasets using **Python**, employing **Matplotlib** and **Power BI** for data visualization, and applied Fourier transformations for frequency analysis and Pearson correlation to uncover underlying data relationships
- Applied **K-means** clustering for anomaly detection and developed custom **feature engineering** pipelines, optimizing the detection of irregular patterns and enhancing the system's responsiveness to network anomalies

ACADEMIC PROJECTS

Comic Book Recommender System

11/2023

- Built a collaborative filtering recommender system using **Python**, **Scikit-learn** and **TensorFlow**, applying data collection, cleaning, and hyperparameter tuning to optimize performance on the Goodreads dataset
- Fine-tuned a **ResNet-50** model on ImageNet to extract visual embeddings, preprocessing comic book covers for high-definition training and integrating PCA-reduced embeddings into a hybrid **recommendation system**

Multimodal Virtual Assistant for Agricultural Workers

01/2022 - 05/2022

- Designed and developed a **Python**-based tool integrating **computer vision** and **NLP** to automate agricultural inquiries using NLTK and SpaCy, providing farm workers with real-time access to critical data and reducing manual research efforts by processing comprehensive datasets of agriculture-related questions from diverse sources
- Fine-tuned **BERT** for enhanced question-response matching accuracy, applying data cleaning and mining techniques to ensure reliable and accurate responses to complex inquiries, significantly improving user satisfaction and query resolution efficiency

Causal Discovery in Time Series Analysis: Oil Prices and Clean Energy Stock Performance

10/2024 - Present

- Examined causal relationships using **Python** between oil price fluctuations and clean energy stock prices, visualized indices trend in time series by historical datasets from S&P Global website, devised policy strategies for clean energy
- Applied causal discovery algorithms (PC, FCI, CD-NOD) to test causal interactions and support portfolio and risk management

Benchmarking Large Language Models Across Specialized Domains and Reasoning Tasks

10/2024 - Present

- Designed a comprehensive benchmarking framework to evaluate model (LLaMA, GPT, LLaVA) performance on reasoning tasks
- Constructed custom datasets using simulators for domain-specific and reasoning challenges, fine-tuned pre-trained LLaVA on these customized datasets and tested performance using BLEU and ROUGE to identify model reliability